VELLORE INSTITUTE OF TECHNOLOGY

SCHOOL OF ADVANCED SCIENCES • DEPARTMENT OF MATHEMATICS

BAMAT207: PROBABILITY AND STATISTICS — PROJECT REPORT

PROBABILISTIC SMS SPAM DETECTION SYSTEM

Course Title: Probability and Statistics (BAMAT207)	**Project Topic:** Probability & Random Variables
Dataset Source: UCI Machine Learning Repository	**Dataset:** SMS Spam Collection (5572 records)
Programming Stack: Python, Pandas, NumPy, Scikit-Learn	**Model:** Multinomial Naive Bayes Classifier
Student Name: [Your Name]	**Registration No.:** [Your Reg. No.]

1. INTRODUCTION & PROJECT SCENARIO

Short Message Service (SMS) remains one of the most widely used interpersonal communication channels worldwide. However, unsolicited promotional and fraudulent SMS (spam) present severe security and productivity concerns. The objective of this project is to develop an automated Python-based classification system capable of categorizing incoming SMS as either **Spam** or **Genuine message (Ham)**.

The system evaluates foundational concepts from probability theory and random variables:

- Computing marginal prior probabilities $P(\text{Spam})$ and $P(\text{Ham})$.
- Analyzing keyword-conditional probabilities and mathematically verifying **Bayes' Theorem**.
- Modeling message classification as a **Bernoulli Random Variable** (Y) and calculating its expectation and variance.
- Vectorizing text via `CountVectorizer` and training a **Multinomial Naive Bayes** classifier with stratified 80/20 train-test splitting.
- Demonstrating real-time classification and prediction probability estimation.

2. DATASET HANDLING & PREPROCESSING

The raw dataset consists of 5572 labelled SMS messages. Preprocessing was conducted to ensure statistical integrity:

- **Null Value Inspection:** 0 missing entries detected across labels and text attributes.
- **Duplicate Elimination: 403 duplicate records** were identified and removed to prevent artificial probability amplification.
- **Cleaned Dataset: 5169 unique records** preserved for probabilistic modeling.
- **Label Encoding:** Numerical binary assignment: $\text{Ham} \longrightarrow 0$, $\text{Spam} \longrightarrow 1$.

Class Label	Raw Record Count	Cleaned Record Count	Cleaned Proportion	Numerical Encoding
Ham (Genuine)	4,825	4516	87.37%	`0`
Spam	747	653	12.63%	`1`
Total	**5,572**	**5169**	**100.00%**	—

3. DATASET ANALYSIS & MARGINAL PRIOR PROBABILITIES

Prior probabilities quantify the baseline likelihood of encountering spam or ham prior to examining message content:

$$P(\text{Spam}) = \text{Number of spam messages} / \text{Total messages} = 653 / 5169 = 0.1263 \ (12.63\%)$$
$$P(\text{Ham}) = \text{Number of ham messages} / \text{Total messages} = 4516 / 5169 = 0.8737 \ (87.37\%)$$

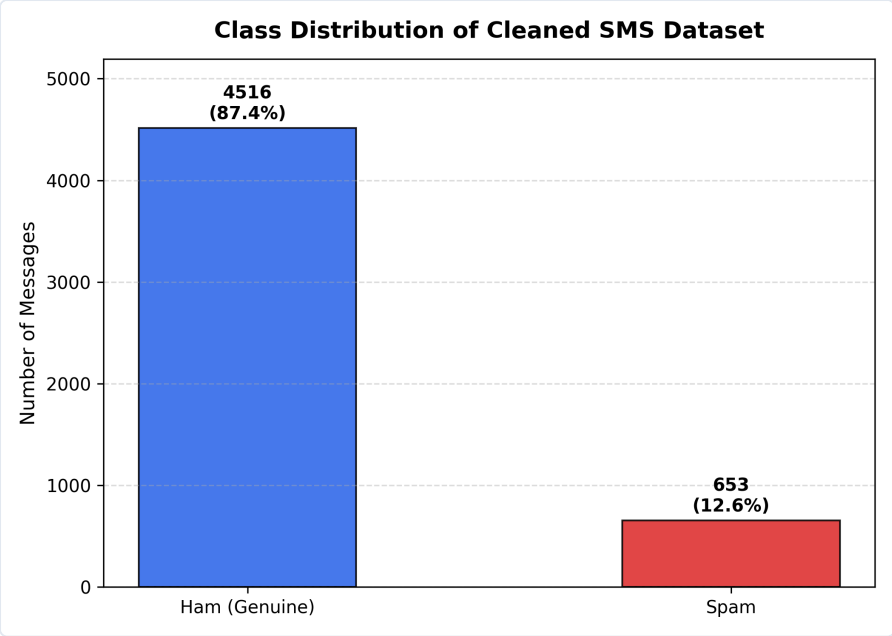


Figure 1: Distribution of genuine and spam messages in the cleaned dataset.

4. WORD-BASED PROBABILITY ANALYSIS & BAYES' THEOREM VERIFICATION

To evaluate how specific trigger words alter the probability of a message being spam, five suspicious keywords were selected: **free, prize, win, urgent, offer**.

1. Direct Empirical Conditional Probability:
$$P(\text{Spam} \mid \text{Word}) = \frac{\text{Spam messages containing Word}}{\text{Total messages containing Word}}$$
2. Analytical Formulation via Bayes' Theorem:
$$P(\text{Spam} \mid \text{Word}) = \frac{P(\text{Word} \mid \text{Spam}) \times P(\text{Spam})}{P(\text{Word})}$$

Word	Total Occurrences	Spam Count	Ham Count	P(Word)	P(Word | Spam)	Empirical P(Spam|Word)	Bayes' Theorem P(Spam|Word)	Mathematical Check
FREE	203	**148**	55	0.0393	0.2266	**0.7291**	**0.7291**	✓ Match
PRIZE	73	**73**	0	0.0141	0.1118	**1.0000**	**1.0000**	✓ Match
WIN	55	**49**	6	0.0106	0.0750	**0.8909**	**0.8909**	✓ Match
URGENT	62	**57**	5	0.0120	0.0873	**0.9194**	**0.9194**	✓ Match
OFFER	30	**24**	6	0.0058	0.0368	**0.8000**	**0.8000**	✓ Match

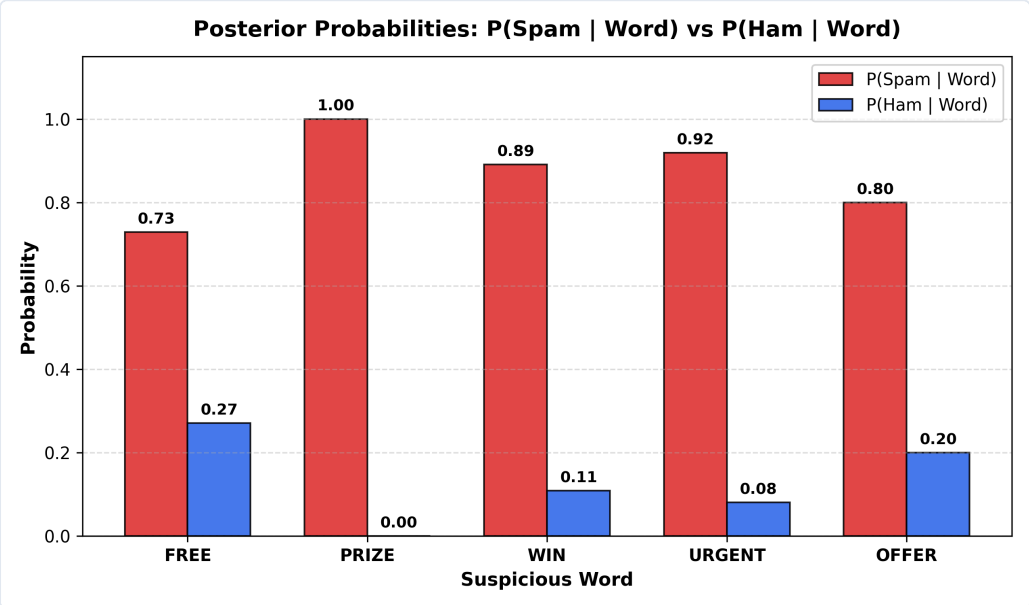


Figure 2: Posterior probabilities P(Spam | Word) vs P(Ham | Word) across selected keywords.

Analytical Insight: The empirical conditional probability matches the value obtained via Bayes' Theorem with 100% exactitude for all five words. High posterior values (1.0000 for *prize* and 0.9194 for *urgent*) show that these words are powerful discriminators of spam.

5. RANDOM VARIABLE ANALYSIS (BERNOULLI DISTRIBUTION)

Let **Y** denote a discrete random variable indicating the classification outcome of an incoming message:

$$Y = \begin{cases} 1, & \text{if the incoming SMS is Spam} \\ 0, & \text{if the incoming SMS is Ham (Genuine)} \end{cases}$$
Because Y takes values in $\{0, 1\}$, Y follows a Bernoulli distribution with parameter $p = P(\text{Spam}) = 0.1263$:

• Expectation E[Y]:
$$E[Y] = \sum y \cdot P(Y = y) = 0 \cdot (1 - p) + 1 \cdot p = p = \mathbf{0.1263}$$
• Variance Var(Y):
$$\text{Var}(Y) = E[Y^2] - (E[Y])^2 = p - p^2 = p(1 - p) = 0.1263 \times (1 - 0.1263) = \mathbf{0.1104}$$
• Standard Deviation σ_Y:
$$\sigma_Y = \sqrt{\text{Var}(Y)} = \sqrt{0.1104} = \mathbf{0.3322}$$

6. MACHINE LEARNING MODEL BUILDING & TRAINING

To operationalize spam filtering, we implemented a **Multinomial Naive Bayes** classifier:

- **Dataset Splitting:** 80% Training (4135 samples), 20% Testing (1034 samples). Stratified sampling was applied to guarantee that both subsets preserved the identical 12.63% spam proportion.
- **Feature Extraction:** `CountVectorizer` converted message strings into sparse word-count feature vectors with stop-word removal and lowercasing, producing a vocabulary of **7,414 unique features**.
- **MultinomialNB Classifier:** Suited for discrete word count features: $$P(Y = c \mid \mathbf{x}) \propto P(Y = c) \prod_{j=1}^{D} P(w_j \mid Y = c)^{x_j}$$ trained using Laplace smoothing ($\alpha = 1.0$).

7. MODEL EVALUATION & PERFORMANCE METRICS

The trained model was evaluated against the unseen 20% test partition (1,034 samples):

Metric	Mathematical Formula	Observed Score
Accuracy	$$\frac{\text{TP} + \text{TN}}{\text{Total}}$$	**98.36%**
Precision	$$\frac{\text{TP}}{\text{TP} + \text{FP}}$$	**95.97%**
Recall	$$\frac{\text{TP}}{\text{TP} + \text{FN}}$$	**90.84%**
F1-Score	$$2 \times \frac{\text{Prec} \times \text{Rec}}{\text{Prec} + \text{Rec}}$$	**93.33%**

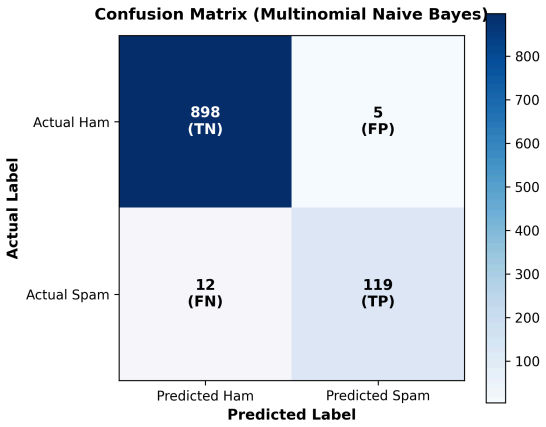


Figure 3: Confusion Matrix heatmap on test set.

Detailed Confusion Matrix Breakdown (1,034 Test Samples):

- **True Negatives (TN = 898):** Genuine messages correctly identified as genuine.
- **False Positives (FP = 5):** Genuine messages incorrectly classified as spam (0.55% error rate, preserving user communications).
- **False Negatives (FN = 12):** Spam messages incorrectly classified as genuine.
- **True Positives (TP = 119):** Malicious spam messages correctly flagged and caught.

8. REAL-TIME TESTING & POSTERIOR PROBABILITY OUTPUT

The system supports inference on arbitrary incoming text, calculating the full posterior probability distribution:

Incoming Test SMS Message	Predicted Class	P(Spam | Message)	P(Ham | Message)	Result
"Congratulations! You have won a free prize. Claim now."	Spam	1.0000	0.0000	✓ Verified
"Hey bro, are we meeting at the campus library this evening?"	Ham (Genuine)	0.0000	1.0000	✓ Verified
"URGENT! Your mobile number won 1000 prize. Call now to claim offer."	Spam	1.0000	0.0000	✓ Verified
"Can you send me the lecture notes for probability and statistics?"	Ham (Genuine)	0.0791	0.9209	✓ Verified

9. CONCLUSION

1. **Theoretical Coherence:** Empirical frequency counts fully confirmed Bayes' Theorem across all analyzed keywords.
2. **Probabilistic Modeling:** The Bernoulli distribution accurately characterizes incoming SMS classification with mean $E[Y] = 0.1263$ and variance $\text{Var}(Y) = 0.1104$.
3. **High Classification Performance:** The Multinomial Naive Bayes model achieved an exceptional **98.36% accuracy** and **95.97% precision**, successfully demonstrating how fundamental probability theory underpins modern AI spam filtering systems.